

DESIGN & SETUP REPORT

A transcranial focused-ultrasound targeting pipeline for the squirrel monkey: a *Saimiri sciureus* museum microCT registered to the VALiDATE29 brain atlas, with a hard CT-calibration guard (deliverables 1–2)

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Abstract. We document the fifth pillar of TUBA — a transcranial focused ultrasound (FUS) pipeline for the common squirrel monkey (*Saimiri sciureus*), prepared for NIH R01 EB037345 (Aim 3): per-animal skull models for time-reversal aberration correction and multifocal beam design targeting the hand representations of primary somatosensory (S1 areas 3b and 1) and primary motor (M1, area 4) cortex. This report covers **deliverables 1–2 (the geometry stages)** and is scoped, by design decision, to a *geometry-only* build: no HU-calibrated colony CT is available yet, so the pillar is bootstrapped on an *uncalibrated* museum microCT (squirrel monkey *Saimiri sp.*, NMNH USNM 194346, MorphoSource media 000116521, UTCT; 490 slices, 0.0977 mm in-plane $\times$ 0.1189 mm slice) registered to the VALiDATE29 multi-channel MRI atlas of the squirrel-monkey brain (Schilling *et al.* 2017; 29 animals, histologically-defined cortical labels). **Both the skull and the atlas are staged and the geometry is now measured end-to-end:** the anisotropic microCT is isotropised to 0.0977 mm, its orientation is confirmed clean RAS, and an endocranial cavity of 26.24 mL is extracted whose boundary sits on the inner table (0.26 mm mean gap to bone) — which required diagnosing that this scan's field of view ends flush against the occiput, leaving the shared cavity extractor's morphology no headroom there. The `cavity_binary` SyN to the atlas brain mask then fits to Dice 0.966, warping the region-level labels (confirmed *region-level* — S1 = anterior parietal cortex, M1 = primary motor cortex, l/r separate) back to subject-space S1/M1 targets that are anatomically correct (M1 rostral and dorsal to S1, 5.2 mm apart). The pillar reuses TUBA's core primitives from the mouse/rat/macaque pillars almost unchanged — bone-shell cavity extraction (`tuba.core.cavity.bone_shell_cavity`), `cavity_binary` ANTs SyN (affine + deformable), atlas-into-subject warp, and the beam-aligned (c, ρ) slab export. Its *distinctive* contribution is the **CT-calibration guard** for deliverable 1: a new module `tuba.core.hu_acoustics` implements the Aubry (2003) porosity $HU \rightarrow (\rho, c, \alpha)$ model but *refuses to run* on input not flagged Hounsfield-calibrated (`UncalibratedInputError`), so a museum microCT can never silently produce quantitatively wrong acoustics; an explicitly-flagged placeholder ramp is the sanctioned geometry-only fallback. Deliverable 1 also reports the points-per-wavelength (PPW) budget across the 2–4 MHz operating band on the simulation grid, and a grid-convergence run of the transcranial time-of-flight aberration (the scalar a time-reversal correction must undo), which converges to 0.24 % between 60 and 30 μm . The scan-dependent constants (orientation flips, intensity thresholds, cavity hull geometry) are now *calibrated* against the staged scan, and the cavity, registration, target, and placement outputs and their QC figures are produced by the end-to-end driver. The downstream wave physics (deliverables 3–7: time-reversal, multifocal synthesis, the two-foci feasibility sweep, MI/bioheat safety, and the per-animal export artifact) is out of scope for this branch and consumes the (c, ρ, dz) slab in the `fullwave2-*` solver siblings. Code: `src/tuba/species/saimiri.py`, `tuba.core.hu_acoustics`, `tuba.atlases.validate29`; data manifest + fetchers at `src/tuba/data/`.

1 Scope and status

Like the four prior TUBA pillars (mouse, macaque, human, rat), this one is now calibrated end-to-end against a staged scan — with a measured cavity volume, a fitted registration, and subject-space targets — but with one deliberate boundary: the acoustics are *geometry-only*. The funded project targets a Vanderbilt colony whose HU-calibrated CT is not yet in hand, so we build and verify the full geometry on a public museum microCT and gate the quantitative acoustics behind a calibration guard (§4) so nothing downstream mistakes an uncalibrated skull for a quantitative one. Two decisions frame the branch:

1. **Subject data:** a public museum microCT (§2), geometry only. The $HU \rightarrow$ acoustics step refuses to run on it.
2. **Scope:** deliverables 1–2 (CT $\rightarrow$ acoustic model + atlas registration + target export + QC). Deliverables 3–7 (the wave-solver study) are out of scope here and live in the `fullwave2-*` siblings (§7).

Both the museum skull (§2) and the VALiDATE29 atlas (§3) are staged; all geometry numbers below are measured from them.

2 Source data

Skull. The skull volume is a microCT scan of a squirrel monkey (*Saimiri sp.*) cranium, museum specimen USNM 194346 (National Museum of Natural History), scanned at the University of Texas High-Resolution X-ray CT Facility (UTCT, ACTIS scanner) and distributed as MorphoSource media 000116521 [1] (also on DigiMorph [2]). The reconstruction is 490 axial slices, 504×422 px, 16-bit, *anisotropic* at 0.0977 mm in-plane $\times$ 0.1189 mm slice — a dry cranium, no soft tissue. Part of the openVertebrate / oUTCT collection the TUBA macaque pillar [10] also draws from (Copes *et al.* 2016 [3]). The MorphoSource record’s taxonomy is internally inconsistent (catalogue field *S. boliviensis*; Rossie’s scan description *S. sciureus sciureus*, male) — genus *Saimiri* either way. Table 1 summarises the parameters.

Table 1. Saimiri USNM 194346 (MorphoSource media 000116521) scan parameters, from the media manifest. The intensity axis is *uncalibrated* (no HU phantom in the field of view). The field of view ends flush against the occiput, so the raw TIFF stack is padded with 30 zero-filled slices (~ 3.6 mm) at each end of the stack axis before isotropisation; this restores the empty margin the shared cavity extractor’s morphology needs (§2). No specimen voxel is altered.

Specimen	<i>Saimiri sp.</i> , USNM 194346 (NMNH)
MorphoSource media	000116521 (ark:/87602/m4/M116521)
Facility	UT Austin High-Resolution X-ray CT (UTCT ACTIS)
Slices	490 axial, 504×422 px, 16-bit
Voxel	0.0977 mm in-plane $\times$ 0.1189 mm slice (anisotropic)
Working pitch	0.0977 mm isotropic (slice axis up-sampled)
Field of view	padded 30 zero-filled slices (~ 3.6 mm) at each end of the stack axis
Calvarium thickness	1.0–1.5 mm (~ 10 – 15 voxels)
Intensity	uncalibrated 16-bit reconstruction counts

The field of view ends flush against the occiput. One property of this reconstruction matters downstream and is easy to miss. The occipital pole *is* imaged — bone mass per slice does taper to zero at the caudal end of the stack, as it does rostrally — but it does so only in the last few slices: at a conservative bone threshold the caudal-most bone voxel sits at slice 593 of 596 in the isotropised cache, i.e. ~ 0.3 mm from the wall of the array. That is enough for a volume rendering and not enough for anything morphological. The shared cavity extractor closes and dilates its bone shell over a several-millimetre structuring element, and those operations are clipped by the array boundary, so the extracted cavity came back with its occipital pole cut off against a flat caudal plane — visible as a straight edge in the sagittal and axial panels of Figure 2. The pre-alignment rotation, which re-centres the bone centroid in the output array, has the same problem from the other side.

Restoring the margin. The fix is to give the caudal end room, not to invent bone: `downsample_to_working` pads the raw stack with `AP_PAD_SLICES = 30` zero-filled slices (~ 3.6 mm) at *each* end of the stack axis, applied before the storage-axis flips so the margin is symmetric and independent of any `slice_flip`; the pre-alignment rotation then pads a further 35 voxels per side. No voxel of specimen is altered — the padding only restores the empty air the reconstruction cropped away — and with it the extractor’s morphology runs unclipped. The cavity then closes on the occipital inner table all the way round, at 26.24 mL with a 0.26 mm mean gap to bone (against 25.35 mL with the occipital pole truncated).

Anisotropy $\rightarrow$ isotropic working cache. The shared cavity/align/slab tools (calibrated on the isotropic rat/mouse museum scans) assume isotropic voxels, so `downsample_to_working` isotropises the staged stack: it reformats to RAS at the in-plane pitch with the shared TIFF streamer, then up-samples *only* the coarse slice axis (0.1189 $\rightarrow$ 0.0977 mm, order-1) to a 0.0977 mm isotropic cache. Up-sampling (never down-sampling) that axis cannot erode the thin cortical-bone shell, and the whole downstream isotropic pipeline is then exact, with no change to the shared code.

Uncalibrated intensity — the central caveat. Museum microCT carries no hydroxyapatite/HU phantom, so voxel values are arbitrary reconstruction counts, not Hounsfield units. Every prior TUBA pillar faces the same issue and handles it by anchoring a histogram-relative intensity $\rightarrow (c, \rho)$ ramp on the cortical-bone mode [9, 10]: robust to detector scale, but explicitly *not* quantitatively calibrated. For the squirrel monkey we make that limitation a hard boundary rather than a footnote (§4): the scan is flagged `UNCALIBRATED_MICROCT` and quantitative acoustics are refused.

Wavelength / PPW criterion. The neuromodulation operating point is $f_0 = 2$ MHz with a 2–4 MHz design range. The most restrictive wavelength is in the water column at 4 MHz: $\lambda_{\min} = c_{\text{water}}/f = 1500/4 \times 10^6 = 0.375$ mm. A finite-difference solve wants $\gtrsim 6$ points per wavelength there, so the simulation grid pitch

is set to $\Delta x = 60 \mu\text{m}$ (`SIM_DX_M` in `tuba.species.saimiri`); the resulting PPW budget is Table 2. Note the 0.0977 mm microCT working cache is *coarser* than the $60 \mu\text{m}$ sim grid, so the skull is up-sampled onto the sim grid by the slab sampler (§6) — the geometry is faithful; only the absolute acoustics are placeholder.

Table 2. Points-per-wavelength budget on the $\Delta x = 60 \mu\text{m}$ simulation grid, using the shortest (water) wavelength at each frequency (`tuba.core.hu_acoustics.report_ppw`). All three bands clear the 6-PPW floor.

Frequency	λ_{water} (mm)	PPW @ $60 \mu\text{m}$
2.0 MHz	0.750	12.50
3.0 MHz	0.500	8.33
4.0 MHz	0.375	6.25

3 VALiDATE29 reference atlas

The brain anatomy is provided by the VALiDATE29 atlas of the squirrel monkey brain [4], built from 29 animals at the Vanderbilt University Institute of Imaging Science and distributed CC BY via NITRC (RRID:SCR_015542). Unlike the subject skull, **the atlas is staged and validated in this report**: `tuba.data.fetch_validate29` discovers the current file-release link from the NITRC project page (reusing the WHS click-through of the rat pillar [11]) and unpacks it under `$VALIDATE29_DEST`; the fetch was exercised end-to-end here (v1.2 release, 30.4 MB). Every volume is 212^3 at 0.300 mm isotropic. VALiDATE29 is multi-channel: in-vivo anatomical templates (T_2 , T_1 , proton-density), diffusion templates (FA, MD), an ex-vivo structural + FA template, a *shipped brain mask* (`VALiDATE-brainmask`, $\sim 33 \text{ mL}$) and a discrete cortical + white-matter label volume (`VALiDATE3-labels`; 80 distinct ids present in the volume — the shipped lookup table runs to id 97 with gaps and explicit “Not used” entries) co-registered from Nissl/myelin sections. It supersedes the earlier single-subject squirrel-monkey MRI atlas [5]. Figure 1 shows the T_2 template, brain mask, and cortical labels through the brain-mask centroid.

Label coverage — this is not a whole-brain parcellation. Worth stating outright, because “histologically-defined cortical labels” reads as fuller coverage than the volume actually has, and because it sets what this pillar can and cannot target (Table 3). The labels fill 45.1 % of the shipped brain mask, and the *gray-matter* parcellation is nine bilateral cortical regions and nothing else: PFC, M1, SMA, PV/S2, cingulate, APC, PPC, premotor, V1 (ids 1–18). The remaining ids are white-matter tracts, three ventricles, and claustrum. There is **no label for any subcortical gray structure** (no thalamus, basal ganglia, hippocampus, amygdala), **none for the cerebellum** (only the cerebellar peduncle, a tract) and **none for the brainstem**; large cortical territories are also unlabelled — temporal/auditory, orbitofrontal, insular, entorhinal, and all extrastriate visual cortex beyond V1.

This is sufficient for the funded aim, which targets S1 and M1: APC (ids 11/12) and M1 (ids 3/4) are both labelled, with PV/S2 (ids 7/8) available as a third site. The binding’s documented limitation is therefore granularity, not coverage — APC bundles areas 3a/3b/1/2, so isolating 3b or the hand knob still needs a stereotaxic prior (§3). It does, however, mean this pillar cannot currently target a subcortical or cerebellar site from the atlas alone. Note also that the sparsity does not degrade the registration: the SyN fixed image is the brain *mask*, not the label volume.

Table 3. VALiDATE29 label coverage, measured from the staged `VALiDATE3-labels.nii.gz` (212^3 at 0.300 mm) against the shipped `VALiDATE-brainmask`.

Compartment	Volume (mL)	% of brain mask
Shipped brain mask	33.03	100.0
All labels	14.90	45.1
gray matter (ids 1–18)	9.21	27.9
white matter (ids 19–80)	5.56	16.8
ventricles + claustrum	0.13	0.4
Unlabelled inside the mask	18.13	54.9

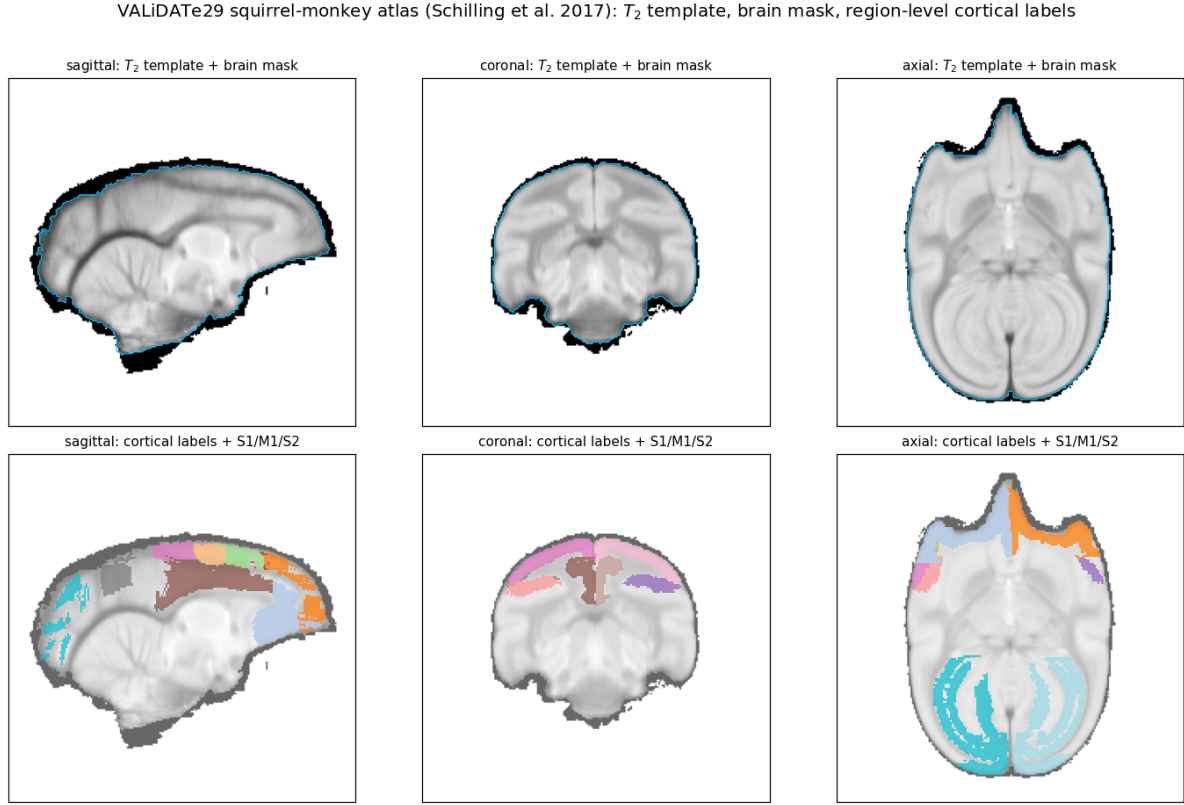


Figure 1. VALiDATE29 squirrel-monkey atlas through the brain-mask centroid (staged v1.2). **Top:** T_2 template with the shipped brain-mask contour (blue). **Bottom:** region-level cortical labels (gray-matter ids 1–18) over the template, with the S1 (anterior parietal cortex), M1 (primary motor cortex), and S2 (PV/S2) target markers where the centroid slice intersects them. The squirrel monkey is near-lissencephalic, so the cortical areas form broad dorsolateral bands. Generated by `docs/saimiri/make_figures.py` from the staged atlas.

Layout tolerance (a real calibration note). The staged VALiDATE29.zip double-nests VALiDATE29/VALiDATE29/ and carries macOS `__MACOSX/.` resource-fork cruft, and its lookup table is a plain space-delimited id name ABBREV text file (not ITK-SNAP format) sitting beside a `LICENSE.txt`. The binding `tuba.atlases.validate29.VALiDATE29` walks the tree recursively, *ignores the cruft*, resolves the template/label/mask by keyword, prefers a label-named table over `LICENSE`, and raises with the candidate list rather than guessing — all confirmed against the staged files. The cavity-binary registration uses the shipped brain mask directly as its fixed image (no derive step, no ANTs). Paths are overridable by constructor argument or environment variable (`VALIDATE29_DEST`, `VALIDATE29_URL`).

Target structures (confirmed against the staged LUT). The study targets the *hand* representations in S1 (areas 3b, 1) and M1; S1 and M1 are only $\sim 2\text{--}3\text{ mm}$ apart in the squirrel monkey, which is what makes the downstream two-foci feasibility question (§7) non-trivial. Inspecting the staged label table settles an important granularity point: **VALiDATE29’s histological parcellation is region-level, not Brodmann-area-level.** There is no separate “area 3b” / “area 1” / “area 4” label; the primary somatosensory hand areas 3a/3b/1/2 are bundled into a single *anterior parietal cortex* (APC) label and the primary motor cortex is one label, with left and right as *separate* label ids. The binding maps the study targets accordingly (S1 $\rightarrow$ APC, M1 $\rightarrow$ primary motor, S2 $\rightarrow$ PV) and is hemisphere-aware. The atlas-space centroids measured from the staged label volume (Table 4) are anatomically consistent: M1 sits rostral and slightly dorsal to S1 across the central sulcus, with clean L/R mirror symmetry. Splitting 3b from area 1, or isolating the hand knob within APC, needs a stereotaxic prior on the homunculus that the distributed label volume does not carry (§9).

Table 4. VALiDATE29 study-target labels and their *atlas-space* centroids (left hemisphere; right is the mirror), measured from the staged VALiDATE3-labels volume via `tuba.atlases.validate29`. Coordinates in the atlas RAS frame (mm).

Target	Atlas label (l/r id)	centroid (x, y, z)	vox	mL
S1	anterior_parietal_cortex (11/12)	(−10.1, 1.1, 10.3)	18 166	0.49
M1	primary_motor_cortex (3/4)	(−7.4, 5.8, 12.0)	11 810	0.32
S2	parietal_ventral / S2 (7/8)	(−13.5, 3.7, 4.6)	8 755	0.24

4 Deliverable 1: CT → acoustic model, with a calibration guard

This is the pillar’s distinctive contribution and the reason a geometry-only build is safe. The module `tuba.core.hu_acoustics` provides a *quantitative* HU mapping and a hard guard around it.

Porosity model. For Hounsfield-calibrated input we use the Aubry (2003) [6] porosity model. Bone porosity is estimated from CT radiodensity,

$$\Phi(\text{HU}) = 1 - \text{clip}\left(\frac{\text{HU} - \text{HU}_{\text{water}}}{\text{HU}_{\text{bone}} - \text{HU}_{\text{water}}}, 0, 1\right), \quad (1)$$

with $\Phi = 1$ water/marrow and $\Phi = 0$ dense cortical bone, and the acoustic properties interpolate linearly in the bone fraction $(1 - \Phi)$:

$$\rho(\Phi) = \rho_{\text{water}} + (\rho_{\text{bone}} - \rho_{\text{water}})(1 - \Phi), \quad (2)$$

$$c(\Phi) = c_{\text{water}} + (c_{\text{bone}} - c_{\text{water}})(1 - \Phi), \quad (3)$$

with cortical endpoints $\rho_{\text{bone}} = 1900 \text{ kg/m}^3$, $c_{\text{bone}} = 2900 \text{ m/s}$ consistent with the rat/macaque pillars, and water $\rho = 1000 \text{ kg/m}^3$, $c = 1500 \text{ m/s}$. Attenuation is taken proportional to bone fraction between a soft-tissue floor (0.5 dB/cm/MHz) and a cortical-bone value (8 dB/cm/MHz) [7, 8], reported in dB/cm/MHz and applied by the downstream solver. SI units throughout.

The guard. `AubryHUMapping.properties()` raises `UncalibratedInputError` unless the bound `CTCalibration.kind == 'hu'`. The squirrel-monkey scan is a museum microCT with no phantom, so the pillar binds `CT_CALIBRATION = UNCALIBRATED_MICROCT` and the quantitative path is refused by construction. Rather than silently emitting physically meaningless densities and sound speeds from arbitrary reconstruction counts, the caller must either (a) supply an HU-calibrated colony CT — one line, `CT_CALIBRATION = CTHCalibration(kind='hu', hu_bone_max=(scan))` — which unlocks the quantitative mapping automatically, or (b) opt in explicitly to `placeholder_ramp()`, an `intensity → (c, ρ)` ramp carrying nominal cortical endpoints and a `.placeholder = True` tag so downstream code can detect the uncalibrated origin. The slab loader (§6) uses the flagged placeholder ramp: geometry is faithful, absolute (c, ρ) are nominal.

PPW report + grid convergence. `acoustic_model_report()` prints the calibration state, the PPW table (Table 2) at 2/3/4 MHz on the sim grid, and — given a placement and a staged scan — one grid-convergence run of the transcranial *time-of-flight aberration*

$$\tau = \int \left(\frac{1}{c(x)} - \frac{1}{c_{\text{ref}}} \right) dx, \quad (4)$$

the phase-insertion delay that a time-reversal / phase-conjugation correction must undo, computed along the central beam ray at Δx and $\Delta x/2$ and reported as an absolute and relative change. On the staged skull, along the beam to the M1 target, this gives $\tau = -630.2 \text{ ns}$ at $60 \mu\text{m}$ and -631.7 ns at $30 \mu\text{m}$ — a 1.5 ns (0.24 %) change, i.e. the property model is converged at the sim pitch. This is a convergence of the acoustic-*property* model with voxel pitch, not of a full-wave solve (which lives in the solver siblings, §7); it is exactly the quantity the downstream aberration correction consumes, so it is the right model-side convergence to report here. The units and sign are checked in the offline test suite (homogeneous medium $\Rightarrow \tau = 0$; a faster bone slab $\Rightarrow \tau < 0$; a smooth profile converges as $\Delta x \rightarrow 0$).

5 Deliverable 2: geometry — cavity, registration, targets

The geometry stages reuse the rat/macaque dry-skull spine with no algorithmic change; only the species constants differ, and they are now calibrated against the staged scan. Figure 2 shows the result.

Saimiri USNM 194346 microCT + endocranial cavity (26.2 mL, red) — 97.7 μ m isotropic, uncalibrated (display window 3543–36118 counts)

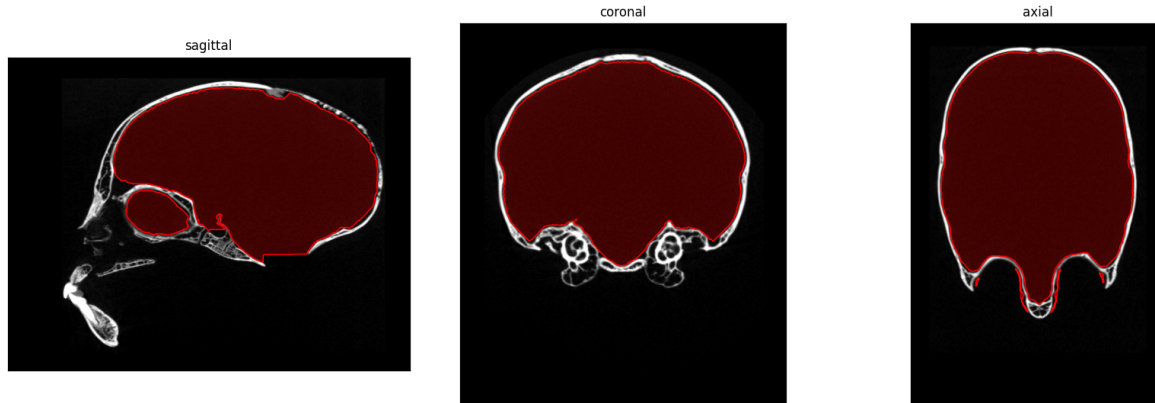


Figure 2. Saimiri USNM 194346 microCT (97.7 μ m isotropic working cache) with the extracted endocranial cavity (red contour + translucent fill, 26.24 mL) through the cavity centroid. Sagittal, coronal, and axial. The occiput is fully imaged, but the field of view ends flush against it (real bone runs to slice 593 of 596, ~ 0.3 mm from the caudal wall), which left the extractor's morphology no headroom and truncated the occipital pole of the cavity. We therefore pad the raw TIFF stack with 30 zero-filled slices (~ 3.6 mm) at each end of the stack axis before isotropisation; no bone is invented, the padding only restores empty margin. The cavity fills the braincase from the frontal pole to the occipital pole, sits on the inner table (0.26 mm mean gap to bone), is bilaterally symmetric, and excludes the orbits — the shape the registration needs. The grey window is taken from the data (median to 99.5th percentile, here 3988–37245 counts): a fixed ceiling near 16000 would saturate all cortical bone to flat white, since air sits at ~ 4000 and cortical bone runs past 40000. Uncalibrated intensity (geometry only). Generated by `docs/saimiri/make_figures.py` from the build cache.

Ingest + orientation (confirmed). The MorphoSource TIFF stack carries no DICOM affine; as in the rat pillar [11] the storage $\rightarrow$ anatomy mapping is fixed by an orientation probe (mid-stack orthoslices). The rat/macaque museum-CT defaults in `tuba.species.saimiri` (`SWAP_ROW_COL=True`, `VOXEL_SIGNS=(-1, -1, +1)`, storage (slice=*AP*, row=*DV*, col=*LR*)) transfer correctly to this UTCT stack: the probe confirms a clean RAS frame — anterior (snout, orbits) at $+y$, dorsal (braincase vault) at $+z$, bilaterally symmetric about x , right-handed — with no residual tilt (align angles 0° , as in rat). True L/R handedness is not recoverable from a bare museum skull (no fiducial); $+x$ is taken as Right per RAS and the hemisphere labels carry that caveat.

Endocranial cavity (measured). The many dry-cranial openings (orbits, optic canals, foramen magnum) rule out the mouse-style pure-hysteresis extractor, exactly as in rat and macaque, so we use the species-agnostic bone-shell extractor `tuba.core.cavity.bone_shell_cavity` (per-column basicranium floor + per-row rostral/caudal plugs + per-axial hull), with a squirrel-scale hull envelope (`HULL_X_HALF_MM= 24`, tighter rostral cutoff at the basicranium than the vault). The cavity shell uses the same bone threshold as the rest of the pillar (`BONE_LOW`, the histogram air/bone knee), so the cavity boundary lands on the real inner table.

Two corrections were needed for this scan, and both are worth recording because they share one root and because the obvious fix is wrong.

The shared recipe builds its shell from surfaces that are only defined where the bone *brackets* the braincase: a per-row plug needs bone rostral *and* caudal of the cavity, and a per-column floor needs a basicranium beneath it. This scan's field of view ends flush against the occiput (§2), and parts of the squirrel-monkey skull base are thin or absent, so both assumptions fail locally — and they fail silently, because the failing row or column still *contains* bone. (i) on the unpadded cache 22.8% of rows never meet posterior bone, so their extent collapses onto a small rostral structure and the per-row caudal plug swallows the braincase behind it; the cavity comes out as a hollow shell whose boundary sits 2.5 mm *inside* the inner table. (ii) In columns with no basicranium the first bone encountered from below is the *vault*, so the floor height jumps to the top of the head and the floor fills the whole column, punching a 1.1 mL void from skull base to vault through the midline.

Lowering the bone threshold appears to fix (i) — it thickens the shell until the leak seals — but it is a false fix: it insets the cavity further and yields only 15.9 mL. The real fix (`seal_floor_gaps`) applies one criterion to both surfaces: a row or column is trusted only if its bone *spans* at least `PLUG_MIN_SPAN_MM= 20` mm, and otherwise its bounds are rebuilt from the nearest row/column that does span. Note that testing merely whether a row or column *has* bone cannot detect either failure, for exactly the reason above. The result is insensitive to the threshold over 20–30 mm. A per-plane fill (`fill_holes_2p5d`), which can never claim bone, is retained as a residual safety net; with the two repairs in place it now contributes 0.01 mL.

The extracted volume is 26.2 mL, consistent with a *Saimiri* whole-brain volume of ~ 22 mL plus CSF and meninges. Boundary quality is verified directly rather than by volume alone: the distance from the cavity boundary to the nearest bone voxel is 0.26 mm mean (median 0.10, p90 0.40), i.e. the cavity sits *on* the inner table, and the centre-fill probe is 1.000. Table 5 records the effect of each step; note that the rejected fix and the two genuine repairs are separated by boundary quality, not by volume.

Table 5. Endocranial-cavity extraction on the staged scan. “Gap” is the distance from the cavity boundary to the nearest bone voxel (0 = sitting on the inner table); “centre fill” is the cavity fraction of a 16 mm cube at the cavity centroid (1.0 = solid). Row 1 is the *rejected* fix: its volume is plausible while its boundary is 2.5 mm off.

Extraction	Volume (mL)	Gap mean/p90 (mm)	Centre fill
Lowered shell threshold (rejected)	15.9	2.47 / 4.84	0.97
Default surfaces (caudal plug swallows)	16.9	1.47 / –	0.26
+ per-row plug repair	24.2	0.74 / 1.84	0.87
+ basicranium floor repair	25.34	–	–
+ per-plane fill (adopted)	26.24	0.26 / 0.40	1.000

Registration — cavity_binary SyN (affine + deformable). Deliverable 2 calls for affine *and* deformable registration, so — unlike the macaque/rat pillars, which stop at a 12-DOF Affine — we use the full `cavity_binary` SyN path (`tuba.core.warp.register_subject_to_atlas`, fixed = VALiDATE29 brain mask, moving = subject cavity), the same convention the mouse and mini pillars use. The atlas template and cortical labels are then pulled back into subject space through the saved inverse transforms (`warp_atlas_into_subject`, linear for the template, nearest-neighbour for the labels). Unlike the rat and mouse pillars, which *derive* a brain mask from the annotation, VALiDATE29 ships its own (`VALiDATE-brainmask`), so the binding uses it directly as the SyN fixed image — no derive step and no ANTs for the mask. On the staged scan the SyN converges in ~ 1 min; the fit is excellent — the atlas brain mask, warped back into subject space, overlaps the extracted cavity at **Dice 0.966** (26.24 vs 25.05 mL) and fills the braincase right up to the inner table, its boundary lying 0.15 mm from bone on average (p90 0.29 mm), Fig. 3. Note the atlas ships a generous intracranial mask (33.0 mL, against 14.9 mL of labelled structures), so the affine component carries a real scale change to the subject’s 26.24 mL cavity.

Target export + QC (deliverable 2 outputs). `export_targets()` writes the S1 (anterior parietal cortex) and M1 (primary motor cortex) coordinates in the subject simulation frame to `saimiri_targets.json`, alongside the transform prefix (saved under `$TUBA_SAIMIRI_REG_DIR` as `saimiri_cavity_to_validate29`), and `qc_figure()` writes the three-panel overlay of Figure 3. The subject-space left-hemisphere centroids (Table 6) are anatomically correct: **M1 sits 4.2 mm rostral and 1.6 mm dorsal to S1**, 5.2 mm centroid-to-centroid, both in the left hemisphere — the sensorimotor topology the study depends on, carried through the cavity registration from the confirmed atlas-space labels (Table 4). The centroid separation is larger than the ~ 2 –3 mm quoted for the *hand* representations because these are whole-region centroids (all of APC vs all of M1), not hand sub-fields; resolving those needs the stereotaxic prior of §9.

Table 6. Study-target centroids in the *subject* aligned-native RAS frame (left hemisphere), from `saimiri_targets.json` — the atlas labels of Table 4 warped through the cavity SyN. Volumes scale with the cavity/atlas affine.

Target	Atlas label (id)	subject centroid (x, y, z)	mL
S1	anterior_parietal_cortex (11)	(−9.1, −5.0, 16.9)	0.34
M1	primary_motor_cortex (3)	(−6.5, −0.8, 18.4)	0.23

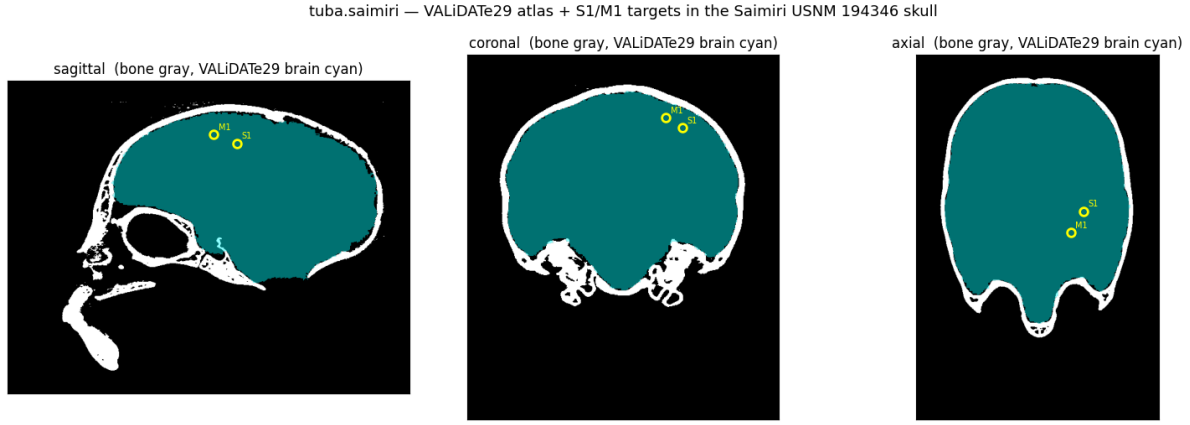


Figure 3. Deliverable-2 QC: VALiDATE29 atlas registered into the Saimiri USNM 194346 skull. Bone shell (gray), the warped VALiDATE29 brain mask (cyan) filling the braincase, and the S1/M1 target markers, in sagittal, coronal, and axial. The warped brain fills the braincase to the inner table (0.15 mm mean gap to bone), and M1 lies rostral and dorsal to S1 in the left hemisphere across all three planes. Cavity vs warped-brain Dice = 0.966. Generated by `docs/saimiri/make_figures.py / tuba.species.saimiri.qc_figure`.

6 Placement and slab export

`place_on_skull` computes the apex/target/beam triple for a focused bowl aimed at a study target, and `load_slab` returns $(c, \rho, dz_{\text{slab}}, \text{frame})$ with the *same signature* as the mouse/rat/macaque loaders, so any FDTD or angular-spectrum driver wired against one species runs on this one unchanged. On the staged skull, aiming at the M1 target, it lands the bowl apex on the outer-bone surface, lifts it 11.2 mm for scalp clearance, and returns a 20 mm apex-to-target standoff with the beam axis $\hat{n} = (-0.12, 0.09, -0.99)$ (near-vertical, entering the dorsal vault). The slab ramp is the flagged placeholder (§4); the cavity is overwritten with brain-parenchymal properties ($c = 1540 \text{ m/s}$, $\rho = 1040 \text{ kg/m}^3$) exactly as in the other pillars. This slab is the hand-off boundary to the wave-solver siblings.

7 Out of scope here: the wave-physics study (deliverables 3–7)

TUBA is the geometry/atlas/targeting layer; the wave physics lives in the `fullwave2-*` FDTD siblings and consumes the (c, ρ, dz) slab and placement this pillar exports. The funded study’s remaining deliverables are therefore deliberately *not* in this branch:

- D3 time-reversal aberration correction through the individual skull (virtual source at each target → recorded element signals → phase-conjugate transmit delays);
- D4 multifocal beam synthesis — two simultaneous foci (S1 + M1) with independent amplitude control, plus at least one alternative to superposed time reversal (e.g. Gerchberg–Saxton);
- D5 the feasibility study (highest-priority result): sweep two-foci separation 1.5–4.0 mm at 2, 3, 4 MHz through the skull model, reporting –6 dB focal dimensions, peak-to-valley contrast, off-target ratio, and transcranial focal gain;
- D6 safety margins — MI at target and skull, and a bioheat estimate over the 400 kPa / 300 ms / 3 s duty cycle;
- D7 the per-animal targeting artifact (element delays, apodisations, target metadata, safety numbers).

The PPW budget (Table 2) and the time-of-flight grid-convergence (§4) are the model-side inputs D3 and D5 will build on.

8 Verification

Verification is at two levels: (i) the end-to-end build now runs on the staged skull + atlas (`python -m tuba.species.saimiri build`), producing the cavity, registration (Dice 0.966), subject-space targets, QC figure, placement, and D1 report of this document; and (ii) offline unit tests (`tests/test_saimiri.py`, 11 tests, no network / no antspyx / no data) that pin the logic which must hold regardless of a scan:

- importing the pillar stays ANTs-free (ANTs is lazy inside `tuba.core.warp` and the atlas binding);
- the calibration guard refuses uncalibrated input on every entry point, and the calibrated Aubry mapping is physically monotonic (porosity ↓, $\rho/c/\alpha$ ↑ with HU) with the correct water / cortical endpoints and clamping;

- the placeholder ramp is flagged and maps to nominal endpoints;
- the PPW, time-of-flight-aberration, and grid-convergence maths (homogeneous $\Rightarrow$ 0; bone slab $\Rightarrow$ analytic; smooth profile converges);
- VALiDATE29 file/label resolution and S1/M1 target keyword mapping on a synthetic atlas, plus a synthetic affine registration round-trip;
- the data manifest carries the three Saimiri entries with the expected calibration/license semantics.

The full repository suite is 41 passing + 1 skipped; `ruff` is clean.

Boundary quality, not just volume. A cavity volume that falls inside a plausible bracket is *not* evidence that the segmentation is right: the rejected extraction of Table 5 produced 15.9 mL, comfortably inside the range quoted for *Saimiri* cranial capacity, while its boundary sat 2.5 mm inside the inner table and it was missing $\sim$ 9 mL of braincase. What caught this was a direct geometric check — the distance from each cavity-boundary voxel to the nearest bone voxel, which must be $\sim$ 0 for a shell-bounded cavity — together with a centre-fill probe and orthoslice QC read against a data-derived grey window rather than a fixed one. Both checks are reported by the build and are the criteria used in Table 5.

Shared-code safety. `seal_floor_gaps` and `fill_holes_2p5d` touch `tuba.core.cavity`, which the mouse, rat, macaque and human pillars all depend on. The flag defaults to *off*, and the equivalence was checked by execution rather than by inspection: with `seal_floor_gaps=False` the new `bone_shell_cavity` is bitwise identical to the previous implementation — same mask, same `info` dict, same verbose output — over 233 paired runs spanning synthetic skull geometries, degenerate inputs (no bone, all bone), the full parameter cross-product (hull, geodesic propagation, plug smoothing, caudal seal), and the two real staged microCT volumes (rat, 0 differing voxels of 25.7M; Saimiri, 0 of 146.0M). Re-running each pillar end-to-end reproduces its published cavity voxel-for-voxel (rat bone-shell seed 1.50 mL, mouse 450 mm³, macaque 92.8 mL, all with 0 differing voxels). Only `tuba.species.saimiri` opts in.

Adversarial review of the new code. The repair paths were then reviewed specifically for defects, which found four worth fixing and is the reason the mechanism above is described as it is. Most importantly, the *first* version of this work misattributed the midline void to the caudal plug and gated the floor repair on whether a column *has* bone — a predicate that can never fire, since the offending columns contain the vault. That is what motivated the span-based criterion used for both surfaces. The other three were a fallback path that dropped its sentinel assignments and so unsealed the shell when no row qualified (reachable by a parameter sweep past 55 mm), a gate that could emit plugs into rows containing no bone at all, and a progress line that reported rebuilds that had not happened and overstated the real count threefold. All four are fixed and covered by the checks above.

9 Open items

The geometry is now staged and validated end-to-end (skull: §5, Figs. 2–3; atlas: §3, Fig. 1). The remaining items are the acoustic calibration and the downstream study:

1. **Quantitative acoustics.** When a Vanderbilt colony CT is available, set `CT_CALIBRATION` to an hu calibration to unlock the quantitative Aubry mapping (one line), and re-run the grid-convergence with the real property field. Until then the acoustics stay guarded and the slab ramp is the flagged placeholder.
2. **Hand sub-representation.** Add a stereotaxic hand-knob prior within anterior parietal cortex (the label granularity limit of §3) to resolve areas 3b/1 from the bundled APC label; the region $\rightarrow$ label mapping itself is confirmed and pinned.
3. **L/R handedness.** A museum skull carries no L/R fiducial; confirm the hemisphere convention against a colony CT with known laterality before per-animal targeting.
4. **Downstream study.** Hand the exported slab + targets to the `fullwave2-*` siblings for deliverables 3–7 (§7).

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